Operating Systems · 21CSC202J

Unit 1 · Session 12 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 12 : SLO â€“ 1 | System Programs vs System Calls â€” Diagram Keywords

I. Diagram Keywords (Layered View)

• Top layer: USER PROGRAMS / APPLICATIONS
• Next: SYSTEM PROGRAMS (utilities: shells, editors, compilers, loaders)
• Next: SYSTEM CALL INTERFACE (API: open, read, write, fork, exec)
• Core: KERNEL (process management, memory management, file system, device drivers)
• Bottom: HARDWARE (CPU, memory, devices)

II. Distinction

• SYSTEM CALLS: the programmatic interface between a process and the kernel (e.g. fork, read).
• SYSTEM PROGRAMS (utilities): executable programs shipped with the OS that USE system calls to
• provide user-facing functionality (e.g. ls, cp, gcc, top). They run in user mode, not kernel mode.